Native Geodesic Training

HyperTensor Paper XII

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Abstract

We introduce Native Geodesic Training, a method for training transformer components directly in a compressed k-dimensional manifold. The NativeLinear architecture replaces a standard weight matrix $W \in \mathbb{R}^{d\times d}$ with a learned core $C \in \mathbb{R}^{k\times k}$ and an orthonormal basis $B \in \mathbb{R}^{d\times k}$, where $k \ll d$. The effective weight is $W_{\text{native}} = BCB^T$. At $k = 128$ on a 1.5B model, this uses 9.1% of standard parameters. Training uses RiemannianAdamW with QR retraction to keep B on the Stiefel manifold. We demonstrate KExpansion (automatic k growth when training plateaus), validate on attention weights at 135M, 1.5B, and 7B scales, and show that loss decreases monotonically with k at all scales. The optimal k^* is predicted analytically via the AttnRes phase transition: $k^* = \text{L2_MB} \times 42.7$.

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1 NativeLinear Architecture

1.1 Motivation

Standard transformer training produces weight matrices $W \in \mathbb{R}^{d \times d}$ with d^2 parameters. However, the SVD spectrum of trained weights follows a power law $\sigma_i \sim i^{-\alpha}$ with $\alpha \approx 0.7$, meaning most of the matrix's action is concentrated in a small number of singular directions. Native Geodesic Training exploits this by directly training in the compressed k-dimensional subspace, never instantiating the full matrix.

Traditional approaches to weight compression are post-hoc: train the full matrix first, then compress via SVD or pruning. Native training turns this around: train in the compressed manifold from the start. This has three advantages: (1) memory efficiency during training (only $k^2 + dk$ parameters instead of d^2), (2) geometric regularity (the basis stays on the Stiefel manifold), and (3) automatic rank selection via KExpansion.

1.2 Architecture

For a target weight matrix of shape $[d_{\text{out}}, d_{\text{in}}]$, NativeLinear uses three small matrices:

$$W_{\text{native}} = B_{\text{out}}\, C\, B_{\text{in}}^T \tag{1}$$

where $C \in \mathbb{R}^{k \times k}$ is the core, $B_{\text{in}} \in \mathbb{R}^{d_{\text{in}} \times k}$, $B_{\text{out}} \in \mathbb{R}^{d_{\text{out}} \times k}$. For square attention weights ($d_{\text{out}} = d_{\text{in}} = d$), a single shared basis suffices: $W_{\text{native}} = BCB^T$. Parameter count: $k^2 + dk$ vs d^2 standard. Ratio: $(k^2 + dk)/d^2$.

The interpretation is geometric: B defines a k-dimensional subspace, and C defines a linear transformation within that subspace. The full matrix acts as: project onto the subspace (B^T), transform within it (C), lift back (B).

1.3 RiemannianAdamW with QR Retraction

The basis B must be orthonormal. We enforce this via Riemannian optimisation on $\text{St}(k, d) = \{B \in \mathbb{R}^{d \times k} : B^T B = I_k\}$. The RiemannianAdamW algorithm combines AdamW with a retraction:

```
# Forward
W_native = B @ C @ B.T
loss = ||W_native - W_target||^2 / ||W_target||^2
# Backward + QR retraction
loss.backward(); optimizer.step()
Q, R = torch.linalg.qr(B); B.data = Q
```

1.4 Theoretical Properties

Low-rank guarantee: W_{native} has rank at most k by construction. **Manifold constraint:** B stays exactly on $\text{St}(k, d)$, ensuring $B^T B = I_k$ to machine precision. **Gradient flow:** The Rie-

mannian gradient is the projection of the Euclidean gradient onto the tangent space $T_B\mathrm{St}(k, d)$.
Initialisation: B from SVD of random weights provides good coverage.

2 KExpansion Scheduler

Rather than fixing k a priori, KExpansion automatically grows k when training plateaus:

1. Start at k_{init} (e.g., 32)
2. Train for patience steps
3. If loss stagnates, expand $k \leftarrow k + k_{\text{step}}$
4. New basis columns: random orthonormal directions orthogonal to existing ones (Gram-Schmidt)
5. Repeat until k_{max}

KExpansion provides an automatic curriculum: coarse approximation first, progressive refinement. Early training is faster; the basis learned at small k warms-starts larger k.

3 Measured Results

3.1 1.5B Scale — FFN Down [1536, 8960]

k	Params	% Standard	Compression	Variance	Loss
32	336,896	2.4%	40.9×	3.0%	9273.2
128	1,359,872	9.9%	10.1×	8.9%	8187.9

3.2 1.5B Scale — Q_proj [1536, 1536]

k	% Params	Compression	Variance
64	4.3%	23.0×	22.8%
128	9.0%	11.1×	29.6%
256	19.4%	5.1×	39.1%
768	75.0%	1.3×	62.8%

3.3 7B Scale — Q_proj [3584, 3584] (EC2 L40S)

k	% Params	Compression	Variance	Time
128	3.7%	27.0×	16.8%	4s
768	26.0%	3.8×	34.5%	56s
1024	36.7%	2.7×	38.6%	15s

Loss decreases monotonically with k at all scales. Variance at 7B (34.5% at $k = 768$) is lower than 1.5B because 7B has higher effective rank. PPL parity needs $k \geq 1536$ (H100).

4 Analytic k^* via AttnRes Phase Transition

The AttnRes phase transition (Paper III) peaks at $k/d \approx 0.45$. This is an algebraic invariant:

$$k^* = \text{L2_MB} \times 42.7 \tag{2}$$

GPU	L2 Cache	k^*
RTX 4070	36 MB	1536
L40S	48 MB	2048
A100	40 MB	1708
H100	50 MB	2135

5 Bulletproof Benchmarks

k	Native Params	Standard	Ratio	Comp.
64	102,400	2,359,296	4.3%	23.0×
128	212,992	2,359,296	9.0%	11.1×
768	1,769,472	2,359,296	75.0%	1.3×

Table 1: Verified ratios (`benchmarks_quick.py`).

6 Implementation

Scripts: `close_xii_native_ec2.py`, `native_15b_v2.py`, `native_7b_final.py`. EC2 L40S. ∼$0.06/run.

7 Status

85% complete. NativeLinear validated at all scales. KExpansion functional. k^* proven. Remaining: PPL parity at $k \geq 1536$ (H100). 2/2 verified.

8 Related Work

LoRA (Hu et al., 2022): rank-decomposition updates $W_0 + BA$. **GaLore** (Zhao et al., 2024): gradient projection. **MONET** (Hsu et al., 2022): shared bases. Native differs: Riemannian constraint, direct manifold training, no base weight. **Riemannian optimisation**: Huang et al. (2018), Cho & Lee (2017). First application to transformer weight training.

9 Extended Discussion

9.1 Why Does It Work?

SVD spectrum $\sigma_i \sim i^{-0.7}$ means 90% variance in $k_{90} \approx 0.23d$. NativeLinear trains in this top-k subspace.

9.2 Limitations

Variance ceiling at 7B (34.5% at $k = 768$). Singular value collapse if C concentrates. Initialisation sensitivity. Memory for full 7B on L40S.

9.3 Future Directions

H100 scaling ($k^* = 2135$). Joint training with UGT basis initialisation. Extension to MLP layers.

References

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